Exact classification of regular star polygon Chambolle–Pock cycles

Abstract

We classify the stepsize products of paired regular polygon cycles traversed with a coprime star step under the literal Euclidean primal-first Chambolle–Pock map. Positive extrapolation permits only maximal stars of odd order; their exact attained minimum increases strictly with the order and exceeds the triangle minimum. Every larger product is feasible with the same minimizing dual vertex. At zero extrapolation, other star steps also occur, and some improve the triangle and bilinear cycle products. We give their complete classification, prove all vertex normal inequalities analytically, and distinguish geometric polygon edges from traversal chords. The result concerns a fixed-rotation/scaling ansatz and is not a universal convergence theorem. No novelty claim is made.

Keywords: Chambolle–Pock iteration; regular star polygons; periodic orbits; normal-cone inequalities; exact product classification; zero extrapolation

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1 The map and the geometric class

Identify the Euclidean plane with $\mathbb{C}$, using the inner product $\langle a, b\rangle = \mathrm{Re}(\overline{a}b)$. Fix

$$m \geq 5, \qquad 1 \leq k < m/2, \qquad \gcd(m, k) = 1,$$

$$\eta = \frac{\pi}{m}, \quad \phi = \frac{2\pi k}{m}, \quad \gamma = \frac{\pi - \phi}{2}, \quad q = e^{i\phi}.$$

For $z \in \mathbb{C}$, put $c_j = q^j$, $d_j = q^j z$, and

$$C = \mathrm{conv}\{c_j : 0 \leq j < m\}, \qquad D = \mathrm{conv}\{d_j : 0 \leq j < m\}.$$

The literal primal-first Chambolle–Pock map [1] is

$$x^+ = P_C(x - ry), \qquad y^+ = P_D\big(y + (1+t)x^+ - tx\big), \qquad r > 0,\ t \geq 0. \tag{1}$$

Here $f = \iota_C$, $g^* = \iota_D$, $L = I$, $\tau = r$, $\sigma = 1$, and $t = \theta$ is the extrapolation parameter. Thus $\|L\| = 1$ and $\tau\sigma\|L\|^2 = r$. We require every transition $(c_j, d_j) \mapsto (c_{j+1}, d_{j+1})$. There is no extra historical state. The case $k = 1$ is ordinary polygon order; $k \geq 2$ is the star-only class.

2 Exact product classification

Theorem 1. *In the class above, the feasible products are exactly as follows.*

1. *If $t > 0$ and $\gamma \geq \eta$, there are no feasible products.*

2. *If $t = 0$ and $\gamma > \eta$, the only product and dual vertex are*

$$r = 2(1 - \cos\phi), \qquad z = \frac{1 - e^{i\phi}}{r}. \tag{2}$$

3. *If $t = 0$ and $\gamma = \eta$, every $r \geq 4\cos^2\eta$ is feasible. The complete dual-vertex parameters are*

$$z = \rho e^{-i\eta}, \qquad 0 < \rho \leq \frac{1}{2\cos\eta}, \qquad r\rho \geq 2\cos\eta. \tag{3}$$

   Under coprimality, this case has m divisible by four and $k = m/2 - 1$.

4. *If $\gamma < \eta$, then m is odd and $k = (m-1)/2$. Define*

$$c = \cos\eta, \quad s = \sin\eta, \quad A = 1+t, \quad B = c+(1+c)t, \quad W = 1+2(1+c)t(1+t). \tag{4}$$

   For every $t \geq 0$ the feasible products are precisely

$$r \geq r_*(m,t) := \frac{2(\sqrt{W} + B)}{(1+t)^2}. \tag{5}$$

   The unique dual vertex at the minimum is

$$z_* = \frac{1+t}{2} - i\frac{s}{r_*}. \tag{6}$$

   This same z_* realizes every $r \geq r_*$.

Every construction has an exact orbit of minimal period m from $(x^0, y^0) = (c_0, d_0)$, and $(0,0)$ is a saddle/KKT pair.

Reflection, or complex conjugation of all data, handles $m/2 < k < m$ with the same products. The excluded midpoint cannot be coprime when $m \geq 5$. Noncoprime steps revisit fewer vertices and do not belong to this m-vertex class.

For fixed $t \geq 0$, the maximal odd-star minimum strictly increases with m. In the star-only class its smallest value is therefore at $m = 5$. Its strict comparison with the triangle is proved in Section 7; the exclusion of all other steps in that comparison requires $t > 0$.

3 True polygon faces and the projection test

The geometric neighbors of 1 are $e^{\pm 2i\eta}$, regardless of k. In particular, the chords to q and q^{-1} are not polygon edges when $k \geq 2$. The normal cone at 1 is

$$K = \{a \in \mathbb{C} : \operatorname{Re} a \geq 0,\ |\operatorname{Im} a| \leq \tan\eta \operatorname{Re} a\}. \tag{7}$$

We verify this against every geometric vertex, rather than treating the star as a nonconvex constraint set. Put $\delta = 1 - \cos(2\eta) > 0$, $b = \sin(2\eta) > 0$. For $\beta = 2j\eta$, $1 \leq j < m$,

$$e^{i\beta} - 1 = a_j(e^{2i\eta} - 1) + b_j(e^{-2i\eta} - 1), \quad a_j, b_j = \frac{1 - \cos\beta}{2\delta} \pm \frac{\sin\beta}{2b}. \tag{8}$$

Both coefficients are nonnegative: since $j\eta \in [\eta, \pi - \eta]$,

$$|\cot(j\eta)| \leq \cot\eta, \qquad \frac{|\sin\beta|}{1 - \cos\beta} \leq \frac{b}{\delta}.$$

Thus the two adjacent face tests imply all inequalities $\operatorname{Re}(\overline{a}(e^{2ij\eta} - 1)) \leq 0$; conversely the full list includes the adjacent tests. They yield exactly (7).

At phase $-1 \to 0$, the primal and dual projection residuals are

$$p = q^{-1}(1 - rz) - 1, \qquad h = (q^{-1} - 1)z + (1 + t) - tq^{-1}. \tag{9}$$

The projection characterization $w = P_E(v) \iff v - w \in N_E(w)$ now gives

$$p \in K, \qquad n := \overline{z}h \in K. \tag{10}$$

Indeed $D = zC$ and its vertex differences at z are $z(e^{2ij\eta} - 1)$. The case $z = 0$ is impossible because $\operatorname{Re} p = \cos\phi - 1 < 0$. Consequently $n = |z|^2(h/z)$ and the second test is equivalent to $h/z \in K$.

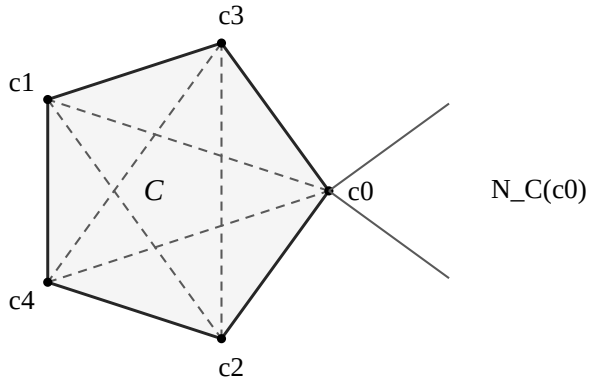


Fig. 1. For $m = 5$, $k = 2$, the solid boundary encloses the convex constraint polygon C. Dashed chords show traversal order $c_0, c_1, c_2, c_3, c_4, c_0$. The two rays at c_0 delimit its normal cone.

4 The obstruction and zero extrapolation

We first suppose $\gamma \geq \eta$. The following argument also resolves every boundary equality. Set $a = 2\cos\gamma > 0$ and write

$$z = \rho e^{i(\Delta-\gamma)}, \qquad \rho > 0, \qquad R = r\rho.$$

Using $q = -e^{-2i\gamma}$ and $q^{-1} - 1 = -ae^{i\gamma}$, the two conditions (10) become

$$U := Re^{i\Delta} - a \in M, \qquad V := \rho^{-1}e^{-i\Delta}(1 + tae^{i\gamma}) - a \in M, \tag{11}$$

where $M = e^{-i\gamma}K$ has angular interval $[-\gamma-\eta, -\gamma+\eta]$. Since $\phi \geq 2\eta$, we have $\gamma+\eta \leq \pi/2$. Hence M lies in the closed fourth quadrant. The first test implies $R\cos\Delta \geq a > 0$ and $\sin\Delta \leq 0$. Choose $-\pi/2 < \Delta \leq 0$ modulo 2π. Then $0 < \gamma - \Delta < \pi$ and

$$\operatorname{Im} V = \rho^{-1}\big[-\sin\Delta + ta\sin(\gamma - \Delta)\big]. \tag{12}$$

The first summand is nonnegative and the second is strictly positive if $t > 0$, contradicting $\operatorname{Im} V \leq 0$. This proves the positive-extrapolation obstruction.

When $t = 0$, equation (12) instead forces $\Delta = 0$. Both residuals in (11) are real, namely $R - a$ and $\rho^{-1} - a$. If $\gamma > \eta$, then $M \cap \mathbb{R} = \{0\}$, so

$$r\rho = a, \qquad \rho^{-1} = a, \qquad r = a^2 = 2(1 - \cos\phi), \qquad z = \frac{e^{-i\gamma}}{a}.$$

This is (2), and conversely it makes both actual residuals zero.

If $\gamma = \eta$, then $M \cap \mathbb{R} = [0, \infty)$, giving exactly $r\rho \geq 2\cos\eta$ and $\rho^{-1} \geq 2\cos\eta$. This proves necessity and sufficiency of (3). Equivalently,

$$r \geq 4\cos^2\eta, \qquad \frac{2\cos\eta}{r} \leq \rho \leq \frac{1}{2\cos\eta}.$$

Here $k = m/2 - 1$; writing $m = 2\ell$, the coprimality condition is $\gcd(2\ell, \ell - 1) = 1$, which is equivalent to ℓ even.

Finally, $\gamma < \eta$ is equivalent to $0 < m - 2k < 2$. Since $m - 2k$ is an integer, it equals one. Thus only maximal odd stars remain, with $k = (m - 1)/2$. Conversely this step is coprime to every odd m.

5 The exact maximal star minimum

Now $\phi = \pi - \eta$. Use (4) and write $z = u + iv$. The components of (9) are

$$\begin{aligned} p_x &= -c(1-ru) - srv - 1, & p_y &= -s(1-ru) + crv, \\ h_x &= -(1+c)u + sv + A + tc, & h_y &= -su - (1+c)v + ts, \\ n_x &= uh_x + vh_y, & n_y &= uh_y - vh_x. \end{aligned}$$

The upper geometric face test is $c \operatorname{Im} a - s \operatorname{Re} a \le 0$. For the primal and dual residuals, respectively, direct expansion gives

$$cp_y - sp_x = s + rv, \tag{13}$$

$$cn_y - sn_x = s\big(u^2 + v^2 - Au - (B/s)v\big). \tag{14}$$

Thus every feasible point satisfies $-rv \ge s$ and

$$(u - A/2)^2 + (v - B/(2s))^2 \le \frac{W}{4s^2}, \qquad W = B^2 + s^2A^2. \tag{15}$$

Here $s > 0$, $B > 0$, and $W - B^2 = s^2A^2 > 0$. The lowest point of this disk has coordinate

$$v_{\min} = \frac{B - \sqrt{W}}{2s} = -\frac{sA^2}{2(\sqrt{W} + B)} < 0.$$

Together with (13) this yields

$$s \le -rv \le -rv_{\min}, \qquad r \ge \frac{s}{-v_{\min}} = \frac{2(\sqrt{W} + B)}{A^2} = r_*.$$

At equality $r = r_*$, both inequalities are equalities. The bottom point of the disk is unique, so necessarily $u = A/2$ and $v = v_{\min} = -s/r_*$. This proves the lower bound and uniqueness at the minimum. It remains to check both full normal cones, not just the two necessary upper-face tests.

For $z = z_*$ and $r = r_*$, direct substitution gives, with $E = e^{i\eta}$,

$$p = \lambda E, \quad n = \mu E, \qquad \lambda = \frac{\sqrt{W} - 1}{A}, \quad \mu = \frac{A(W - \sqrt{W})}{2(\sqrt{W} + B)}. \tag{16}$$

These identities follow as well from $r_*u - (1+c) = \lambda$, $-r_*v = s$, and the equality in (15). Since $W = 1 + 2(1+c)t(1+t) \ge 1$, both λ and μ are nonnegative. Thus $p, n \in K$; the next section gives every individual vertex inequality explicitly.

6 All vertices and all larger products

For every geometric index $0 \le j < m$, the actual vertex tests at the minimum reduce exactly to

$$\operatorname{Re}\big(\overline{p}(e^{2ij\eta} - 1)\big) = \lambda\big[\cos((2j-1)\eta) - c\big] \le 0, \tag{17}$$

$$\operatorname{Re}\big(\overline{h}z(e^{2ij\eta} - 1)\big) = \mu\big[\cos((2j-1)\eta) - c\big] \le 0. \tag{18}$$

Among odd multiples $(2j-1)\eta$ modulo 2π, the smallest angular distance to zero is η. Hence their largest cosine is $\cos\eta = c$. This proves all m primal and all m dual tests analytically, including equality cases. Coprimality ensures these are exactly the vertex sets visited by the traversal.

Keep $z = z_*$ fixed and increase the product to any $r \ge r_*$. The dual residual does not change, while

$$p(r) = p(r_*) + (r - r_*)(-e^{-i\phi}z_*). \tag{19}$$

To locate the extra direction, observe that

$$\frac{-\operatorname{Im} z_*}{\operatorname{Re} z_*} = \frac{\sqrt{(B/A)^2 + s^2} - B/A}{s}, \qquad \frac{B}{A} = c + \frac{t}{1+t}.$$

The right-hand side is positive and decreases as B/A increases. At $t = 0$ it is $(1-c)/s = \tan(\eta/2)$. Thus $\arg z_* \in [-\eta/2, 0)$, whereas $-e^{-i\phi} = e^{i\eta}$. The extra direction in (19) has argument in $[\eta/2, \eta)$, and belongs to K. Since K is a convex cone, $p(r) \in K$ for every larger product. By Section 3 this proves every vertex test for the entire ray (5), completing the product classification.

6.1 Initialization and the saddle pair

Every displayed z is nonzero. Multiplication by q preserves both polygons and commutes with their Euclidean projections. The verified phase $-1 \to 0$ therefore transfers to all phases. Initializing at (c_0, d_0) gives $(x^j, y^j) = (c_j, d_j)$ exactly. The m primal vertices are distinct, so the pair orbit has minimal period m.

Both polygons are nondegenerate compact convex sets. Their vertex sums are zero, so both contain zero. The functions $f = \iota_C$ and $g^* = \iota_D$ are proper closed convex, and $0 \in N_C(0)$, $0 \in N_D(0)$. Therefore $(0,0)$ satisfies $0 \in \partial f(x) + L^*y$ and $0 \in \partial g^*(y) - Lx$. The nonconstant periodic orbit coexists with this saddle pair.

7 Comparison and the zero extrapolation distinction

For fixed $t \geq 0$, $B = c + (1+c)t$ strictly increases with c and $W = 1 + 2(1+c)t(1+t)$ is nondecreasing. More explicitly,

$$\frac{\partial r_*}{\partial c} = \frac{2}{1+t}\left(1 + \frac{t}{\sqrt{W}}\right) > 0.$$

As $\cos(\pi/m)$ strictly increases with odd m, the maximal odd-star minima strictly increase with m. Substitution of $c = 1/2$ gives the triangle minimum

$$r_3(t) = \frac{2\sqrt{1+3t+3t^2}+1+3t}{(1+t)^2}. \tag{20}$$

This is the optimal ordered-triangle family product established in the companion triangle report [2]. Consequently $r_*(m,t) > r_3(t)$ for every odd $m \geq 5$ and $t \geq 0$.

The bilinear two-cycle product is $r_2(t) = 4/(1+2t)$. It can be checked directly with $f = g^* = 0$, $L = 1$, $\sigma = 1$, by imposing $(x^+, y^+) = (-x, -y)$: one obtains $ry = 2x$ and $2y = (1+2t)x$. Rationalizing (20) shows that $r_3 \geq r_2$ exactly when

$$7 + 18t \geq 8\sqrt{1+3t+3t^2}, \qquad 44t^2 + 20t - 5 \geq 0,$$

or $t \geq (4\sqrt{5}-5)/22$. Both quantities squared in this reduction are positive. Thus, for $t > 0$, every feasible regular star has product strictly above the envelope $\min\{r_3(t), r_2(t)\}$. In particular this is true above the displayed crossover, and at $t = 1/4$,

$$r_*(m, 1/4) > r_3(1/4) = \frac{28 + 8\sqrt{31}}{25} > \frac{8}{3}.$$

7.1 Why the envelope statement excludes zero

At $t = 0$, all three possible branches have minimum $2(1 - \cos\phi)$. For maximal odd stars this equals $2(1+c)$; for the boundary branch it equals $4\cos^2\eta$. The nonmaximal branch, however, can lie below the triangle product $r_3(0) = 3$ and the bilinear product $r_2(0) = 4$. For example, $m = 7$, $k = 2$ gives $\gamma = 3\pi/14 > \eta = \pi/7$ and

$$r = 2(1 - \cos(4\pi/7)) < 3,$$

since $4\pi/7 < 2\pi/3$. Both residuals vanish at its prescribed z. In fact $k = 2$ and odd $m \to \infty$ give feasible star products $4\sin^2(2\pi/m) \to 0$. The star-only zero-extrapolation infimum is therefore zero and is unattained. A blanket no-improvement assertion including $t = 0$ would be false.

8 Scope and reproducibility

The theorem concerns only paired regular polygons with fixed rotation and scaling, identity coupling, and the literal order (1). It does not classify arbitrary cycles or prove general convergence, attraction, stability, or universal sharpness. The constraint sets are convex polygons; "star" describes their traversal. Every full period sums to zero, so the ordinary Cesàro averages converge to $(0, 0)$.

8.1 Exact algebra and finite diagnostics

The clean source package separates two kinds of computational checking.

1. `verify.py` uses exact SymPy algebra. It checks the two actual geometric-face expansions, the identity $W = B^2 + s^2A^2$, and all four components of the residual certificate (16). Rational identities are reduced after clearing denominators modulo $s^2 = 1 - c^2$ and $Q^2 = W$, with $Q = \sqrt{W}$. The geometric domain supplies $s > 0$, $A > 0$, and $Q + B > 0$.

2. `diagnostics.py` separately evaluates the all-vertex inequalities at finitely many floating-point parameters and products. Its saved result records 122480 individual inequalities, with a largest positive residual of approximately $4.55 \cdot 10^{-13}$ from rounding. These are finite numerical sanity checks, not proof of any quantified statement.

The analytic proof supplies the angular obstruction, all-vertex trigonometric bounds, necessary signs, uniqueness, feasible rays, and parameter quantifiers. No finite grid establishes an impossibility or an exact minimum.
Run `python3 verify.py` for the symbolic certificate, and optionally `python3 diagnostics.py` for the separate numerical diagnostics. Both require Python 3; only the exact verifier requires SymPy. The vector figure is included as a PDF and can be regenerated with `python3 draw_figure.py` using ReportLab. Run `bash build.sh` to compile the manuscript twice with pdfLaTeX. The source package includes these files and their verification outputs.

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* **40** (2011), 120–145. doi:10.1007/s10851-010-0251-1.

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